

# Vectors: Components, Magnitude, and Unit Vectors

Pre-Calculus / Calculus Worksheet · Grade 10–12

Name: \_\_\_\_\_

Date: \_\_\_\_\_

## Learning Objectives

- Find the component form of a vector given an initial and terminal point
- Calculate the magnitude (length) of a vector
- Find the unit vector in the direction of a given vector
- Perform vector operations including scalar multiplication and vector addition

## Problems

1. Find the component form of the vector with the given initial point P and terminal point Q.

$$P = (1, 3), \quad Q = (5, 7)$$

2. Find the component form of the vector with the given initial point P and terminal point Q.

$$P = (4, -1), \quad Q = (-2, 6)$$

3. Find the magnitude of the vector below.

$$\vec{v} = \langle 3, 4 \rangle$$

4. Find the magnitude of the vector with initial point P and terminal point Q.

$$P = (0, -3), \quad Q = (5, 9)$$

5. Find both the component form and magnitude of the vector with the initial point P and terminal point Q.

$$P = (3, -2), \quad Q = (-1, 1)$$



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6. Find the unit vector in the direction of the given vector.

$$\vec{V} = \langle 3, 4 \rangle$$

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7. Find the unit vector in the direction of the given vector.

$$\vec{V} = \langle -5, 12 \rangle$$

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8. Given vectors  $V$  and  $W$  below, find  $2V$ .

$$\vec{V} = \langle -2, 5 \rangle, \quad \vec{W} = \langle 3, 4 \rangle$$

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9. Given vectors  $V$  and  $W$  below, find  $W$  minus  $V$ .

$$\vec{V} = \langle -2, 5 \rangle, \quad \vec{W} = \langle 3, 4 \rangle$$

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10. Given vectors  $V$  and  $W$  below, find  $V$  plus  $W$  and then find the magnitude of the resulting vector.

$$\vec{V} = \langle -2, 5 \rangle, \quad \vec{W} = \langle 3, 4 \rangle$$

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# Vectors: Components, Magnitude, and Unit Vectors — Answer Key

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## Answer Key

### 1. Answer: $\langle 4, 4 \rangle$

- Component form =  $(Q_1 - P_1, Q_2 - P_2)$
- $Q_1 - P_1 = 5 - 1 = 4$
- $Q_2 - P_2 = 7 - 3 = 4$
- Component form:  $(4, 4)$

### 2. Answer: $\langle -6, 7 \rangle$

- Component form =  $(Q_1 - P_1, Q_2 - P_2)$
- $Q_1 - P_1 = -2 - 4 = -6$
- $Q_2 - P_2 = 6 - (-1) = 7$
- Component form:  $(-6, 7)$

### 3. Answer: 5

- Magnitude =  $\sqrt{v_1^2 + v_2^2}$
- $= \sqrt{3^2 + 4^2}$
- $= \sqrt{9 + 16}$
- $= \sqrt{25} = 5$

### 4. Answer: 13

- Component form =  $(5 - 0, 9 - (-3)) = (5, 12)$
- Magnitude =  $\sqrt{5^2 + 12^2}$
- $= \sqrt{25 + 144}$
- $= \sqrt{169} = 13$

### 5. Answer: Component: $(-4, 3)$ , Magnitude: 5

- Component form =  $(-1 - 3, 1 - (-2)) = (-4, 3)$
- Magnitude =  $\sqrt{(-4)^2 + 3^2}$
- $= \sqrt{16 + 9}$
- $= \sqrt{25} = 5$

### 6. Answer: $\langle \frac{3}{5}, \frac{4}{5} \rangle$

- Find the magnitude:  $|v| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$
- Unit vector =  $v / |v| = (3, 4) / 5$
- Unit vector =  $(3/5, 4/5)$

### 7. Answer: $\langle -\frac{5}{13}, \frac{12}{13} \rangle$

- Find the magnitude:  $|v| = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$



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- Unit vector =  $v / |v| = (-5, 12) / 13$
  - Unit vector =  $(-5/13, 12/13)$
- 

**8. Answer:  $\langle -4, 10 \rangle$** 

- Scalar multiplication:  $2V = 2 * (-2, 5)$
  - $= (2*(-2), 2*5)$
  - $= (-4, 10)$
- 

**9. Answer:  $\langle 5, -1 \rangle$** 

- $W - V = (3 - (-2), 4 - 5)$
  - $= (3 + 2, 4 - 5)$
  - $= (5, -1)$
- 

**10. Answer:  $V + W = \langle 1, 9 \rangle$ ,  $|V + W| = \sqrt{82}$** 

- $V + W = (-2 + 3, 5 + 4) = (1, 9)$
  - Magnitude =  $\sqrt{1^2 + 9^2}$
  - $= \sqrt{1 + 81}$
  - $= \sqrt{82}$
- 

